

fstab

The **fstab(5)** (<https://man.archlinux.org/man/fstab.5>) file can be used to define how disk partitions, various other block devices, or remote file systems should be mounted into the file system.

Each file system is described in a separate line. These definitions will be converted into **systemd** mount units dynamically at boot, and when the configuration of the system manager is reloaded. The default setup will automatically **fsck** and mount file systems before starting services that need them to be mounted. For example, systemd automatically makes sure that remote file system mounts like **NFS** or **Samba** are only started after the network has been set up. Therefore, local and remote file system mounts specified in `/etc/fstab` should work out-of-the-box. See **systemd.mount(5)** (<https://man.archlinux.org/man/systemd.mount.5>) for details.

The **mount** command will use `fstab`, if just one of either directory or device is given, to fill in the value for the other parameter. When doing so, mount options which are listed in `fstab` will also be used.

1 Usage

A simple `/etc/fstab` , using file system UUIDs:

`/etc/fstab`

```
# <device>                <dir> <type> <options>
UUID=0a3407de-014b-458b-b5c1-848e92a327a3 /      ext4 defaults
UUID=CBB6-24F2             /boot  vfat defaults,nodev,nosuid,noexec,fmask=0177,dmask=0077
UUID=f9fe0b69-a280-415d-a03a-a32752370dee none  swap defaults
UUID=b411dc99-f0a0-4c87-9e05-184977be8539 /home  ext4 defaults
```

- `<device>` describes the block special device or remote file system to be mounted; see **#Identifying file systems**.
- `<dir>` is the directory where the file system will be **mounted** to, a.k.a. the mountpoint. The directory must be created beforehand.
- `<type>` is the **file system** type.
- `<options>` are the file system mount options; see **mount(8) § FILESYSTEM-INDEPENDENT MOUNT OPTIONS** (https://man.archlinux.org/man/mount.8#FILESYSTEM-INDEPENDENT_MOUNT_OPTIONS) and **ext4(5) § Mount options for ext4** (https://man.archlinux.org/man/ext4.5#Mount_options_for_ext4).
- `<dump>` is checked by the **dump(8)** (<https://linux.die.net/man/8/dump>) utility. This field is usually set to `0` , which disables the check.
- `<fsck>` sets the order for file system checks at boot time; see **fsck(8)** (<https://man.archlinux.org/man/fsck.8>). For the root device it should be `1` . For other

Related articles

[Persistent block device naming](#)

[File systems](#)

[tmpfs](#)

[swap](#)

[genfstab](#)

partitions it should be `2` , or `0` to disable checking.

Tip

- The `auto` type lets the `mount` command guess what type of file system is used. This is useful for **optical media** (CD/DVD/Blu-ray).
- If the root file system is **btrfs** or **XFS**, the `fsck` order should be set to `0` instead of `1` . See **fsck.btrfs(8)** (<https://man.archlinux.org/man/fsck.btrfs.8>) and **fsck.xfs(8)** (<https://man.archlinux.org/man/fsck.xfs.8>).

All specified devices within `/etc/fstab` will be automatically mounted on startup and when the `-a` flag is used with **mount(8)** (<https://man.archlinux.org/man/mount.8>) unless the `noauto` option is specified. Devices that are listed and not present will result in an error unless the `nofail` option is used.

See **fstab(5) § DESCRIPTION** (<https://man.archlinux.org/man/fstab.5#DESCRIPTION>) for details.

2 Identifying file systems

There are different ways to identify file systems that will be mounted in `/etc/fstab`: kernel name descriptor, file system label and UUID, and GPT partition label and UUID for GPT disks. Kernel name descriptors should not be used, while UUIDs or PARTUUIDs should be preferred over labels. See **Persistent block device naming** for more explanations. It is recommended to read that article first before continuing with this article.

In this section, we will describe how to mount file systems using all the mount methods available via examples. The output of the commands `lsblk -f` and `blkid` used in the following examples are available in the article **Persistent block device naming**.

2.1 Kernel name descriptors

Run `lsblk -f` to list the partitions and prefix the values in the *NAME* column with `/dev/` .

`/etc/fstab`

#	<device	<dir>	<type>	<options>	<dump>	<fsck>
	/dev/sda2	/	ext4	defaults	0	1
	/dev/sda1	/boot	vfat	defaults,nodev,nosuid,noexec,fmask=0177,dmask=0077	0	2
	/dev/sda3	/home	ext4	defaults	0	2
	/dev/sda4	none	swap	defaults	0	0

Warning

Kernel name descriptors for block devices are not **persistent** and can change each boot, they should not be used in configuration files (including `/etc/fstab`).

2.2 File system labels

Run `lsblk -f` to list the partitions, and prefix the values in the [LABEL](#) column with `LABEL=` or alternatively run `blkid` and use the LABEL values without the quotes:

```
/etc/fstab
```

# <device>	<dir>	<type>	<options>	<dump>	<fsck>
LABEL=System	/	ext4	defaults	0	1
LABEL=ESP	/boot	vfat	defaults,nodev,nosuid,noexec,fmask=0177,dmask=0077	0	2
LABEL=Data	/home	ext4	defaults	0	2
LABEL=Swap	none	swap	defaults	0	0

Note

If any of your fields contains spaces, see [#Filepath spaces](#).

2.3 File system UUIDs

Run `lsblk -f` to list the partitions, and prefix the values in the [UUID](#) column with `UUID=` or alternatively run `blkid` and use the UUID values without the quotes:

```
/etc/fstab
```

# <device>	<dir>	<type>	<options>
UUID=0a3407de-014b-458b-b5c1-848e92a327a3	/	ext4	defaults
UUID=CBB6-24F2	/boot	vfat	defaults,nodev,nosuid,noexec,fmask=0177,dmask=0077
UUID=b411dc99-f0a0-4c87-9e05-184977be8539	/home	ext4	defaults
UUID=f9fe0b69-a280-415d-a03a-a32752370dee	none	swap	defaults

2.4 GPT partition labels

Run `blkid` to list the partitions, and use the [PARTLABEL](#) values without the quotes:

```
/etc/fstab
```

# <device>	<dir>	<type>	<options>	<dump>
PARTLABEL=GNU/Linux	/	ext4	defaults	0
PARTLABEL=EFI\040system\040partition	/boot	vfat	defaults,nodev,nosuid,noexec,fmask=0177,dmask=0077	0
PARTLABEL=Home	/home	ext4	defaults	0
PARTLABEL=Swap	none	swap	defaults	0

Note

If any of your fields contains spaces, see [#Filepath spaces](#).

2.5 GPT partition UUIDs

Run `blkid` to list the partitions, and use the [PARTUUID](#) values without the quotes:

```
/etc/fstab
```

# <device>	<dir>	<type>	<options>
PARTUUID=98a81274-10f7-40db-872a-03df048df366	/	ext4	defaults

```
PARTUUID=d0d0d110-0a71-4ed6-936a-304969ea36af /boot vfat defaults,nodev,nosuid,noexec,fmask=0177,dmask=0177
PARTUUID=7280201c-fc5d-40f2-a9b2-466611d3d49e /home ext4 defaults
PARTUUID=039b6c1c-7553-4455-9537-1befbc9fbc5b none swap defaults
```

3 Tips and tricks

3.1 Automount with systemd

See [systemd.mount\(5\)](https://man.archlinux.org/man/systemd.mount.5) (<https://man.archlinux.org/man/systemd.mount.5>) for all systemd mount options.

3.1.1 Local partition

In case of a large partition, it may be more efficient to allow services that do not depend on it to start while it is checked by *fsck*. This can be achieved by adding the following options to the `/etc/fstab` entry of the partition:

```
x-systemd.automount
```

This will *fsck* and mount the partition only when it is first accessed, and the kernel will buffer all file access to it until it is ready. This method can be relevant if one has, for example, a significantly large `/home` partition.

Note

This will make the file system type `autofs` which is ignored by [locate](#) by default.

3.1.2 Remote file system

The same applies to remote file system mounts. If you want them to be mounted only upon access, you will need to use the `x-systemd.automount` parameters. In addition, you can use the `x-systemd.mount-timeout=` option to specify how long systemd should wait for the mount command to finish. Also, the `_netdev` option ensures systemd understands that the mount is network dependent and order it after the network is online.

```
x-systemd.automount,x-systemd.mount-timeout=30,_netdev
```

3.1.3 Encrypted file system

If you have secondary encrypted file systems with keyfiles, you can also add the `nofail` parameter to the corresponding entries in `/etc/crypttab` and `/etc/fstab`. *systemd* will not wait for the `cryptsetup` service to finish unlocking and mounting the filesystem on boot, but instead may finish mounting this after reaching `default.target`. This will avoid any boot delay caused by unlocking secondary partitions that are not required immediately after boot. See [dm-crypt/System configuration#Non blocking mounting](#) for `cryptsetup` configuration

Since mount services will by default only wait for 90 seconds for the partition to be available, any delay

in making the keyfile available may cause the mount to fail. To avoid this, add the option `x-systemd.mount-timeout=0` to `fstab` in order to make sure that the mount service waits indefinitely for the partition to be unlocked.

```
/etc/fstab
```

```
UUID=0a3407de-014b-458b-b5c1-848e92a327a3 /data ext4 defaults,nofail,x-systemd.device-timeout=0 0 2
```

3.1.4 Automatic unmount

You may also specify an idle timeout for a mount with the `x-systemd.idle-timeout` flag. For example:

```
x-systemd.automount,x-systemd.idle-timeout=1min
```

This will make `systemd` unmount the mount after it has been idle for 1 minute.

3.2 External devices

External devices that are to be mounted when present but ignored if absent may require the `nofail` option. This prevents errors being reported at boot. For example:

```
/etc/fstab
```

```
LABEL=MyExternalDrive /media/backup jfs nofail,x-systemd.device-timeout=5 0 2
```

The `nofail` option is best combined with the `x-systemd.device-timeout` option. This is because the default device timeout is 90 seconds, so a disconnected external device with only `nofail` will make your boot take 90 seconds longer, unless you reconfigure the timeout as shown. Make sure not to set the timeout to 0, as this translates to infinite timeout.

3.3 Filepath spaces

Since spaces are used in `fstab` to delimit fields, if any field (*PARTLABEL*, *LABEL* or the mount point) contains spaces, these spaces must be replaced by escape characters `\` followed by the 3 digit octal code `040` :

```
/etc/fstab
```

```
UUID=47FA-4071 /home/username/Camera\040Pictures vfat defaults 0 0  
LABEL=Storage\040drive /media/100\040GB\040(Storage) ext4 defaults,user 0 2
```

3.4 atime options

Below atime options can impact drive performance.

- The `strictatime` option updates the access time of the files every time they are accessed. This is more purposeful when Linux is used for servers; it does not have much value for desktop use. The drawback about the `strictatime` option is that

even reading a file from the page cache (reading from memory instead of the drive) will still result in a write.

- The `noatime` option fully disables writing file access times to the drive every time you read a file. This works well for almost all applications, except for those that need to know if a file has been read since the last time it was modified. The write time information to a file will continue to be updated anytime the file is written to with this option enabled.
- The `nodirtime` option disables the writing of file access times only for directories while other files still get access times written.

Note

`noatime` implies `nodirtime`. [You do not need to specify both \(https://lwn.net/Articles/244941/\)](https://lwn.net/Articles/244941/).

- `relatime` updates the access time only if the previous access time was earlier than the current modify or change time. In addition, since Linux 2.6.30, the access time is always updated if the previous access time was more than 24 hours old. This option is used when the `defaults` option, `atime` option (which means to use the kernel default, which is `relatime`; see [mount\(8\)](https://man.archlinux.org/man/mount.8) (<https://man.archlinux.org/man/mount.8>) and [Once upon atime \(https://lwn.net/Articles/244829/\)](https://lwn.net/Articles/244829/)) or no options at all are specified.

When using [Mutt](#) or other applications that need to know if a file has been read since the last time it was modified, the `noatime` option should not be used; using the `relatime` option is acceptable and still provides a performance improvement.

Since kernel 4.0 there is another related option:

- `lazytime` reduces writes to disk by maintaining changes to inode timestamps (access, modification and creation times) only in memory. The on-disk timestamps are updated only when either (1) the file inode needs to be updated for some change unrelated to file timestamps, (2) a sync to disk occurs, (3) an undeleted inode is evicted from memory or (4) if more than 24 hours passed since the last time the in-memory copy was written to disk.

Warning

In the event of a system crash, the access and modification times on disk might be out of date by up to 24 hours.

Note that the `lazytime` option works **in combination** with the aforementioned `*atime` options, not as an alternative. That is `relatime` by default, but can be even `strictatime` with the same or less cost of disk writes as the plain `relatime` option.

3.5 Remounting the root partition

If for some reason the root partition has been improperly mounted read only, remount the root partition with read-write access with the following command:

```
# mount -o remount,rw /
```

3.6 GPT partition automounting

When using UEFI/GPT, it is possible to omit certain partitions from `/etc/fstab` by partitioning according to the [Discoverable Partitions Specification \(https://uapi-group.org/specifications/specs/discoverable_partitions_specification/\)](https://uapi-group.org/specifications/specs/discoverable_partitions_specification/) and have [systemd-gpt-auto-generator\(8\)](http://man.archlinux.org/man/systemd-gpt-auto-generator.8) (<http://man.archlinux.org/man/systemd-gpt-auto-generator.8>) mount the partitions. See [systemd#GPT partition automounting](#).

To specify custom mount options for these volumes, use a [by-designator identifier](#) as the device name:

```
/etc/fstab
```

```
/dev/disk/by-designator/root / ext4 defaults,noatime 0 1
/dev/disk/by-designator/swap none swap defaults,discard 0 0
```

3.7 Bind mount

You can link directories with the `bind` option:

```
/etc/fstab
```

# <device>	<dir>	<type>	<options>	<dump>	<fsck>
>					
UUID=94649E22649E06E0	/media/user/OS/	ntfs	defaults,rw,errors=remoun		
t-ro 0 0					
/media/user/OS/Users/user/Music/	/home/user/Music/	none	defaults,bind 0	0	
/media/user/OS/Users/user/Pictures/	/home/user/Pictures/	none	defaults,bind 0	0	
/media/user/OS/Users/user/Videos/	/home/user/Videos/	none	defaults,bind 0	0	
/media/user/OS/Users/user/Downloads/	/home/user/Downloads/	none	defaults,bind 0	0	
/media/user/OS/Users/user/Documents/	/home/user/Documents/	none	defaults,bind 0	0	
/media/user/OS/Users/user/projects/	/home/user/projects/windows/	none	defaults,bind 0	0	

See [mount\(8\) § Bind mount operation \(https://man.archlinux.org/man/mount.8#Bind_mount_operation\)](https://man.archlinux.org/man/mount.8#Bind_mount_operation) for details.

3.8 Automatically generate an fstab using genfstab

You can use the `genfstab` tool to create an fstab file. See [genfstab](#) for details.

3.9 GUI utilities

Here is a list of programs that can be used to modify mount points. They might not have all the features possible for editing fstab, but have all of the most used ones and might make your workflow much easier:

- [GNOME Disks](#) — A GNOME utility for dealing with storage devices. Part of [gnome \(https://archlinux.org/groups/x86_64/gnome/\)](https://archlinux.org/groups/x86_64/gnome/).

<https://apps.gnome.org/DiskUtility/> | | [gnome-disk-utility \(https://archl](#)

[inux.org/packages/?name=gnome-disk-utility](https://archlinux.org/packages/?name=gnome-disk-utility))

- **KDE Partition Manager** — Utility to help you manage the disks, partitions, and file systems on your computer. Part of **kde-system** (https://archlinux.org/groups/x86_64/kde-system/).

<https://apps.kde.org/partitionmanager/> || **partitionmanager** (<https://archlinux.org/packages/?name=partitionmanager>)

3.10 Modifying user permissions and ownership

If you want to allow any user to mount the drive, consider adding these mount options to add onto your fstab entries.

Note

This is also used for filesystems that do not have file permission abilities, which will also set the owner and permissions of the entire drive to the user who mounted it.

- **users** - Allow any user to mount and to unmount the filesystem, even when some other ordinary user mounted it. This option implies the options **noexec**, **nosuid**, and **nodev** (unless overridden by subsequent options, as in the option line **users,exec,dev,suid**). Simply add **users** to the mount options to enable this.
- **user** - Allow an ordinary user to mount the filesystem. Only allows the same user to unmount. This option implies the options **noexec**, **nosuid**, and **nodev** (unless overridden by subsequent options, as in the option line **user,exec,dev,suid**). Simply add **user** to the mount options to enable this.

For filesystems that do not have file permissions built in such as FAT and exFAT, you can explicitly set the user or group for the entire drive and its files. You can view the ID of a specific user in **/etc/passwd**. The uid is the third number in the entry, and the group id is the fourth.

- **uid** - Set the owner ID of the drive
- **gid** - Set the group ID of the drive

For ext4, btrfs, and other filesystems that have permission abilities, other users might not be permitted to see the drive. Be sure to double check the permissions of **/path/to/drive/** and modify them for what you need.

3.11 Verify changes

Use **findmnt --verify --verbose** to check for syntax errors and invalid options in fstab.

4 See also

- **Full device listing including block device** (<https://docs.kernel.org/admin-guide/devices.html>)

- **Filesystem Hierarchy Standard** (https://refspecs.linuxfoundation.org/FHS_3.0/fhs/index.html)
- **30x Faster Cache and Site Speed with TMPFS** (<https://www.askapache.com/optimize/super-speed-secrets/>)

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